



HOMETASK : 05

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SECTION :24_AIT_KRG-G1-B

UID :24BAI70008

SUBJECT CODE :24CSP-310

SUBJECT NAME :Advanced Database Management

System

FACULTY'S NAME : Mr. Alok Kumar

[A.]

Julia asked her students to create some coding challenges. Write a query to print the *hacker_id*, *name*, and the total number of challenges created by each student. Sort your results by the total number of challenges in descending order. If more than one student created the same number of challenges, then sort the result by *hacker_id*. If more than one student created the same number of challenges and the count is less than the maximum number of challenges created, then exclude those students from the result.

Input Format

The following tables contain challenge data:

- *Hackers*: The *hacker_id* is the id of the hacker, and *name* is the name of the hacker.

Column	Type
<i>hacker_id</i>	Integer
<i>name</i>	String

- _____

Challenges: The *challenge_id* is the id of the challenge, and *hacker_id* is the id of the student who created the challenge.

Column	Type
<i>challenge_id</i>	Integer
<i>hacker_id</i>	Integer

ANS:.

SELECT

h.hacker_id,

h.name,

COUNT(c.challenge_id) AS total_challenges

```

FROM Hackers h
JOIN Challenges c
    ON h.hacker_id = c.hacker_id
GROUP BY h.hacker_id, h.name
HAVING COUNT(c.challenge_id) = (
    SELECT MAX(challenge_count)
    FROM (
        SELECT COUNT(*) AS challenge_count
        FROM Challenges
        GROUP BY hacker_id
    ) x
)
OR COUNT(c.challenge_id) IN (
    SELECT challenge_count
    FROM (
        SELECT COUNT(*) AS challenge_count
        FROM Challenges
        GROUP BY hacker_id
    ) x
    GROUP BY challenge_count
    HAVING COUNT(*) = 1
)
ORDER BY total_challenges DESC, h.hacker_id;

```

[B.]

You did such a great job helping Julia with her last coding contest challenge that she wants you to work on this one, too!

The total score of a hacker is the sum of their maximum scores for all of the challenges. Write a query to print the *hacker_id*, *name*, and total score of the hackers ordered by the descending score. If more than one hacker achieved the same total score, then sort the result by ascending *hacker_id*. Exclude all hackers with a total score of 0 from your result.

Input Format

The following tables contain contest data:

- *Hackers*: The *hacker_id* is the id of the hacker, and *name* is the name of

Column	Type
<i>hacker_id</i>	Integer
<i>name</i>	String

the hacker.

- *Submissions*: The *submission_id* is the id of the submission, *hacker_id* is the id of the hacker who made the submission, *challenge_id* is the id of the challenge for which the submission belongs to, and *score* is the score

Column	Type
<i>submission_id</i>	Integer
<i>hacker_id</i>	Integer
<i>challenge_id</i>	Integer
<i>score</i>	Integer

of the submission.

ANS:

SELECT

h.hacker_id,

h.name,

SUM(x.max_score) AS total_score

FROM Hackers h

JOIN (

```
SELECT
    hacker_id,
    challenge_id,
    MAX(score) AS max_score
FROM Submissions
GROUP BY hacker_id, challenge_id
) x
ON h.hacker_id = x.hacker_id
GROUP BY h.hacker_id, h.name
HAVING SUM(x.max_score) > 0
ORDER BY total_score DESC, h.hacker_id;
```